
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Reinforcement learning (RL) has demonstrated substantial empirical effectiveness
2 in improving the reasoning capabilities of language models, but the training dynam-
3 ics by which RL modifies LLM reasoning behavior remain insufficiently specified.
4 This work explains the RL training process through coordinated empirical analysis
5 and formal theoretical modeling. First, using systematic reasoning-pattern-level
6 and token-level measurements throughout training, we show that individual reason-
7 ing patterns maintain relatively stable success rates, whereas RL primarily changes
8 the probabilities of a sparse subset of critical tokens. These localized probability
9 updates alter the distribution over reasoning patterns and thereby affect aggregate
10 model performance. Building on these observations, we develop a theoretical
11 framework for RL training dynamics under two representative reward mechanisms:
12 verifiable reward (RLVR) and the model’s internal feedback (RLIF). For RLVR,
13 we analyze two regimes: one in which the policy rapidly converges to optimal
14 reasoning strategies, and another in which optimization becomes difficult because
15 the base model has weaker initial reasoning quality. For RLIF, we characterize
16 how internal rewards can produce early performance improvements while also
17 allowing degradation under continued training. Extensive experiments validate the
18 proposed analysis and provide technically grounded guidance for applying RL in
19 language-model post-training.

20 1 Introduction

21 Recent frontier reasoning models, including Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1 [14],
22 attain strong performance on structured logical tasks such as mathematical problem solving [30, 44,
23 41] and program synthesis [49, 40, 8]. Reinforcement learning (RL) constitutes a central post-training
24 mechanism in these systems, and its empirical effect is to substantially modify the operational
25 capability profile of an already trained language model.

26 This empirical success has motivated mechanistic and theoretical analyses of RL post-training for
27 LLMs. Using pass@k comparisons, Yue et al. [42] show that RL-post-trained models often fail
28 to exceed the coverage of their base models, which suggests that RL may not introduce entirely
29 new reasoning patterns. From an entropy-based perspective, Cui et al. [9], Zhang et al. [45] prove
30 that RL-based objectives can reduce policy entropy, and Cui et al. [9] further establish an empirical
31 association between policy entropy and model performance. In parallel, Wang et al. [33], Huan
32 et al. [17] show that RL updates concentrate on a sparse subset of high-impact tokens. Despite these
33 analyses, the training dynamics by which RL modifies reasoning behavior remain only partially
34 characterized, both at the empirical level and in formal models.

35 We therefore begin with a systematic analysis of RL training at two complementary granularities:
36 reasoning patterns and individual tokens. At the reasoning-pattern level, we extract and classify
37 model responses using LLM-based and rule-based procedures, then measure the distribution of these

patterns and their associated success rates throughout training. At the token level, we quantify how next-token probability ranks change across checkpoints. Relative to prior analyses [17, 42], these experiments provide more direct evidence that RL predominantly optimizes a sparse set of critical tokens and, through these local probability changes, *reshapes reasoning pattern distributions* in a way that affects aggregate task performance. We also observe that the intrinsic success rate of each individual pattern remains comparatively stable during training. These observations motivate a mathematical abstraction of RL training as distributional reweighting over pre-existing reasoning patterns.

On the basis of these empirical findings, we formulate a two-stage model of reasoning as a question–reason–answer process $q \rightarrow r \rightarrow a$. The first stage selects a reasoning pattern conditional on the question, represented by $\pi(r|q)$, and the second stage generates an answer conditional on both the selected pattern and the question, represented by $\pi(a|r, q)$. Within this framework, we analyze RL training dynamics under two representative reward mechanisms: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, we show convergence toward the reasoning pattern r^* with maximal success rate, matching the empirical pattern-selection behavior. We further distinguish two convergence regimes: models with strong initial reasoning quality converge rapidly toward optimal patterns, whereas weaker models encounter optimization difficulty in an entanglement stage. For RLIF, we explain how internal rewards can produce early performance gains while also permitting convergence to a state whose performance is below that of the base model. Additional case studies and simulations support these theoretical conclusions.

The main contributions of this paper are as follows:

- We provide a systematic empirical characterization of RL training dynamics at the reasoning-pattern and token levels. By combining next-token rank-shift measurements with LLM-based and rule-based pattern analysis, we obtain sharper evidence than prior work [17, 42] that RL updates concentrate on a sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* while leaving the intrinsic success rates of individual patterns relatively stable.
- We introduce a formal two-stage framework that represents reasoning as $q \rightarrow r \rightarrow a$ and use it to analyze two representative RL post-training objectives. For RLVR, we prove convergence to the reasoning pattern with the highest success rate and characterize rapid-convergence and entanglement-stage regimes. For RLIF, we formalize the source of early performance improvements and explain why continued optimization can yield worse performance than the base model.
- We validate the theoretical predictions through additional case studies and numerical simulations. The resulting framework connects empirical observations to explicit training-dynamic mechanisms and provides practical guidance for analyzing and improving RL-based LLM post-training.

2 Related Works

Reinforcement Learning for LLMs Reinforcement learning is a widely used post-training procedure for large language models (LLMs), with demonstrated utility for preference alignment [25, 48] and for high-precision reasoning domains such as mathematics and programming [30, 20, 22]. In conventional RLHF pipelines, the reward component is typically implemented as a learned model trained on human-annotated preference data [26, 25, 2]. More recent systems instead employ reward signals that are cheaper to obtain or directly computable, including reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]. This shift in reward specification has also motivated new optimization algorithms for LLM post-training, including GRPO [30], DAPO [41], and VAPO [43].

Theoretical Analyses and Mechanism Interpretation of RL for LLMs. The empirical effectiveness of RL for LLMs has led to a substantial literature on formal analysis and mechanism interpretation. Early work primarily studied high-level design questions, including reward construction [28, 18, 38] and algorithmic variants for training [37, 11, 21]. As LLM post-training has moved toward simpler reward structures such as RLVR and RLIF, subsequent analyses have examined why these objectives can improve reasoning behavior. Shao et al. [29] argue that even spurious rewards can improve reasoning by activating latent skills acquired during pretraining, whereas Agarwal et al. [4] attribute performance gains to entropy minimization. Gandhi et al. [13] identify specific cognitive behaviors, including backtracking, that correlate with improved reasoning capability. Related work by [46] shows

that RL fine-tuning amplifies pre-trained behaviors and can drive convergence toward dominant output formats; Wang et al. [34], Cui et al. [10] further analyze this phenomenon as entropy collapse. In contrast to these perspectives, our work models RL training dynamics explicitly as reasoning-pattern selection and supports this interpretation with both empirical measurements and a mathematical training-dynamic analysis.

3 Preliminaries

Reinforcement Learning for LLMs Let π_θ be a language model with parameters θ , which serves as the policy to be optimized. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy π_θ generates an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$. The optimization objective can be formulated as:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

where π_{ref} is the base reference policy, and β is a hyperparameter that controls the KL divergence to prevent excessive deviation between π_θ and π_{ref} .

The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ can be implemented in various formats, such as a trained reward model [25] or a rule-based scoring function [30]. In this paper, we focus on verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4].

In RLVR, the reward function $r_\phi(\mathbf{x}, \mathbf{y})$ directly evaluates whether the answer $\mathbf{y}$ matches the correct answer to question $\mathbf{x}$. A typical reward function is defined as:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

In contrast, RLIF [47, 4] defines rewards using only intrinsic signals computed from the model itself. Following Zhao et al. [47], we instantiate the RLIF reward as the negative average KL divergence between the uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s conditional next-token distribution:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

The subsequent sections analyze how these two reward mechanisms induce distinct empirical and theoretical training dynamics.

4 Experimental Exploration for RL Training

We characterize the RL training process through a sequence of controlled empirical analyses under different reward specifications. We first compare the full training trajectories and observe that RLVR produces monotonic and stable improvements, whereas RLIF exhibits instability and can degrade model performance. We then analyze reasoning patterns and find that RLVR increases the probability mass assigned to patterns with higher empirical success rates, while RLIF does not reliably specialize toward those patterns. Finally, a token-level analysis shows that these pattern-distribution changes are induced by rank changes for a small subset of high-impact tokens. Relative to prior work [17, 42], these experiments provide direct evidence that RL updates are sparse at the token level and operate on performance primarily by *reshaping reasoning pattern distributions*.

4.1 Training Procedure Overview for RLVR and RLIF

We first examine complete training trajectories for RLVR and RLIF under matched experimental conditions. Specifically, we use Qwen2.5-3B [39] as the base policy and train on the MATH dataset [16], holding all non-reward settings fixed across the two algorithms.

Figure 1a shows that the two reward mechanisms induce qualitatively different optimization trajectories. Under RLVR, task accuracy increases steadily and approaches a convergent regime. Under RLIF, accuracy can improve during early updates but may decrease under continued optimization, in some runs falling below the original base-model performance. This divergence motivates a finer-grained analysis of the internal behavioral variables that change during RL training.

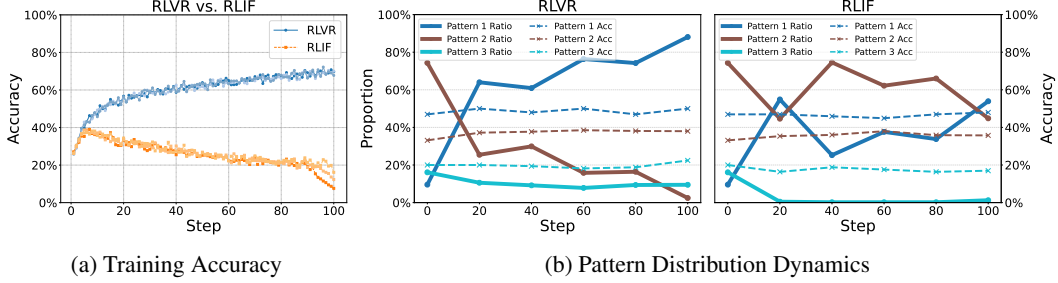


Figure 1: **(a)** Training trajectories for RLVR and RLIF on the MATH dataset. RLVR produces stable accuracy gains, whereas RLIF first increases and then decreases model performance. Each training paradigm is evaluated over multiple random seeds. **(b)** Reasoning-pattern dynamics. The solid curves report the response-level proportion of each pattern among all identified patterns (left axis), and the dotted curves report the corresponding pattern accuracy (right axis). During RLVR, probability mass shifts toward higher-accuracy patterns and away from lower-accuracy patterns; RLIF instead exhibits unstable pattern-distribution dynamics. Patterns are ordered from high to low according to their associated accuracy for visual clarity.

4.2 Reasoning-pattern Level Analysis

To explain the performance gap between RLVR and RLIF, we analyze model behavior at the level of reasoning patterns. We first collect base-model responses and use GPT-4o [19] to cluster responses into pattern categories according to lexical cues and logical structure, while also generating a natural-language description for each category. These pattern assignments are then used to estimate pattern distributions and pattern-specific success rates at each stage of RL training.

We instantiate this analysis on Number Theory problems sampled from the MATH dataset [16] and track both pattern frequencies and pattern accuracies during RLVR and RLIF training. Figure 1b reports the main results, and Appendix A provides additional evaluations across models and datasets. The analysis yields three observations:

- RLVR-trained models consistently shift probability mass toward reasoning patterns with higher success rates, which accounts for the observed steady improvement in aggregate accuracy.
- RLIF induces unstable reasoning-pattern distributions and does not consistently concentrate probability mass on more effective patterns.
- For both reward mechanisms, the success rate associated with any fixed reasoning pattern remains approximately stable over the training process.

These results indicate that the performance gap between RLVR and RLIF is mediated by different dynamics over the reasoning-pattern distribution. We next analyze token-level probability changes to identify the mechanism that produces these distributional shifts.

4.3 Token-level Analysis

Motivated by Huan et al. [17], Yue et al. [42], we further examine the token-level mechanism underlying the observed reasoning-pattern dynamics. We sample base-model responses for questions from GSM8K [6], MATH [16], and AIME [7]. For each fixed response, we compute the rank of each generated token under both the base model and the RL-enhanced model, where rank denotes the position of the token probability among all vocabulary tokens conditioned on the preceding context. Table 1 shows that token ranks are largely invariant: in most settings, fewer than 10% of tokens undergo a rank shift. Thus, RL modifies the probabilities of a sparse set of critical tokens while leaving most token-level choices in the reasoning trace unchanged.

Table 1: Token ranking-change ratio after RL training.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

Together, these experiments support the following mechanistic characterization of RL training dynamics: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains relatively stable during training.*

These empirical findings specify the behavioral variables that the theoretical framework must model: sparse token-level probability changes, redistribution over reasoning patterns, and approximately stable pattern-specific success rates.

5 Theoretical Characterization of RL Training Dynamics

We now translate the empirical observations into a formal model of RL training dynamics. The framework abstracts model reasoning as a two-stage procedure: the policy first selects a reasoning pattern and then generates an answer conditional on that pattern. Within this abstraction, RL primarily acts on the pattern-selection distribution. We use the framework to analyze the effects of different reward mechanisms. For RLVR, we characterize training dynamics in two regimes with distinct convergence behavior. For RLIF, we show how internal rewards can increase performance at initialization while also permitting convergence to policies whose accuracy is below that of the base model.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. Given a question q , we represent reasoning as a structured stochastic process with two components: (1) selection of a reasoning pattern r from the candidate set $\mathcal{R} = \{r_1, r_2, \dots\}$ and (2) generation of a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ conditioned on the selected pattern. The policy selects pattern r_i with probability $p(r_i|q)$, and each pattern is associated with its own success rate. Under this representation, the RLIF objective in Eq. 3.3 can be written as follows¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

Policy Parameterization. Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set, and let $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ denote the set of output sequences. Given a question q , the language model defines an autoregressive distribution over output sequences $y \in \mathcal{Y}$:

$$\text{(general policy)} \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is parameterized by θ and maps the preceding context $y_{<l}$ to logits for the l -th token. The next-token distribution is $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ with $z = f_\theta(x, y_{<l})$.

Because practical neural policies are high-dimensional and nonconvex, deriving optimization guarantees for RL training dynamics is technically difficult [3, 23]. Following prior theoretical analyses [27, 24, 10, 45], we therefore study a *tabular policy* parameterization. This policy is a special case of Eq. 5.2 in which each possible next token is assigned a trainable logit conditional on the preceding token; for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$\text{(tabular policy)} \quad \pi_\theta(y_l|x, y_{<l}) = \pi_\theta(y_l|y_{l-1}) = \text{softmax}(\theta_{:,y_{l-1}})_{y_l}, \quad (5.3)$$

where $\theta_{:,y_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ denotes the column of θ indexed by the preceding token y_{l-1} .

Optimization for RL. To analyze training dynamics, we combine the tabular policy in Eq. 5.3 with the RL objective $\phi_{\text{RL}}(\theta)$ in Eq. 3.1. We study the small-learning-rate limit of policy-gradient optimization through the following gradient flow:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

where $\theta(t)$ denotes the parameters of the policy $\pi_{\theta(t)}$ at training time t , with initialization $\theta(0) = \theta_{\text{ref}}$.

The experiments indicate that the success rate of a fixed reasoning pattern is approximately stable during training. We model this observation by assuming that optimizing the mapping from questions to reasoning patterns is substantially easier than changing the conditional mapping from a selected reasoning pattern to the final answer. This leads to the following assumption:

Assumption 5.1. *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer $\mathbf{r}^*$, defined as $\pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

For the remainder of the section, we impose Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$ for the fixed success rate of pattern $\mathbf{r}$. We first derive the optimal policy for the RL objective in Eq.3.1 under the general autoregressive policy in Eq.5.2. We then analyze RLVR and RLIF training dynamics under the tabular policy in Eq. 5.3.

5.2 The Optimal Policy for RL

The empirical results show that RLVR improves performance by increasing the probability of higher-success reasoning patterns, whereas RLIF produces less stable performance changes. We first characterize the optimal policy induced by the RL objective under these two reward definitions:

Proposition 5.2. *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the reasoning path reward. Specifically:

- For RLVR, $R(\mathbf{r})$ is the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- For RLIF, $R(\mathbf{r})$ is the confidence score induced by the final-answer distribution for reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

Proposition 5.2 identifies the optimal pattern-selection distribution for RLVR and RLIF. Provided that π_{opt} is representable in the autoregressive policy class, which is true for both LLMs and the tabular policies considered below, RLVR increases the probability of a reasoning pattern $\mathbf{r}$ when its success rate $p^*(\mathbf{r})$ is large enough that $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$. By contrast, $R_{\text{RLIF}}(\mathbf{r})$ depends only on confidence in the final-answer distribution and does not distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. Thus, the RLVR optimum is aligned with higher task accuracy, whereas the RLIF optimum does not necessarily improve accuracy relative to the base policy.

Moreover, because β is typically small in practice (e.g., 0.001) [26, 5, 44], the limiting behavior as $\beta \rightarrow 0$ is:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

In this limit, the optimal policy becomes deterministic and always selects the reasoning pattern $\mathbf{r}$ that maximizes $R(\mathbf{r})$, independent of the initial reference distribution over patterns. This optimal-policy characterization does not by itself explain the nonconvex training trajectory: in particular, it does not specify when RLVR reaches high-reward patterns or why RLIF can improve performance early in training. We therefore analyze the gradient-flow dynamics of both methods under the tabular policy in Eq. 5.3.

5.3 Training Dynamic Analysis for RLVR

The experiments indicate that the effectiveness of RLVR depends strongly on the quality of the base model. We therefore analyze the probability assigned to the *optimal reasoning pattern*, defined as the pattern $\mathbf{r}^*$ with the highest success rate for producing the correct answer $\mathbf{a}^*$, namely $\mathbf{r}^* =$

244 $\arg \max_{\mathbf{r}} p^*(\mathbf{r})$. The gradient-flow dynamics exhibit two regimes. In the first regime, the probability
 245 $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically and converges to 1. In the second regime, the policy enters an
 246 entanglement stage in which a suboptimal pattern $\mathbf{r}' \neq \mathbf{r}^*$ impedes optimization toward $\mathbf{r}^*$. After
 247 this stage, the dynamics transition to the first regime and the policy eventually concentrates on $\mathbf{r}^*$.

248 **Theorem 5.3** (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq*
 249 *3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1*
 250 *holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$,*
 251 *defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})p^*(\mathbf{r})$, satisfies:*

$$(Regime 1) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

252 *then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.*

253 Theorem 5.3 formalizes the favorable regime in which the reference model is sufficiently strong: its
 254 overall accuracy exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7). Under
 255 this condition, RLVR drives selection probability toward the optimal pattern at rate $\mathcal{O}(1/\epsilon)$, and the
 256 resulting overall accuracy approaches the success rate of that optimal pattern.

257 In practice, RLVR can be difficult to optimize in settings with weaker reference models [44, 36].
 258 We capture this behavior with a second regime in which the model initially enters an entanglement
 259 stage. In this stage, the *suboptimal reasoning pattern with the second-highest success rate*, defined as
 260 $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, slows the increase of probability mass on the optimal pattern:

261 **Theorem 5.4** (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq*
 262 *3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1*
 263 *holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of*
 264 *the initialized model satisfies:*

$$(Regime 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

265 *then there exists:*

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})} \quad (5.9)$$

266 *with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can*
 267 *guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} >$*
 268 *$p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.*

269 Theorem 5.4 studies a regime in which only the optimal pattern and the second-best pattern have
 270 success rates above the model’s overall accuracy. This restriction simplifies the analysis; the same
 271 mechanism should extend to more general settings in which *at least* two patterns exceed the average
 272 success rate, although the corresponding proof would require a more complex case analysis.

273 The main implication of Theorem 5.4 is that RLVR may require T_0 time steps before the overall
 274 success rate exceeds the success rate of the suboptimal pattern, thereby entering the favorable regime
 275 of Theorem 5.3. Although Theorem 5.3 gives a convergence time T_1 that is polynomial in $1/\epsilon$, the
 276 transition time T_0 in Theorem 5.4 can grow super-exponentially in $\gamma_{\pi_{\text{ref}}}$, the ratio between the total
 277 reference probability assigned to $\mathcal{R}/\{\mathbf{r}'\}$ and the reference probability assigned to $\mathbf{r}^*$. When the base
 278 model assigns very small probability to the optimal pattern, $\gamma_{\pi_{\text{ref}}}$ can be large and T_0 can become
 279 prohibitively long. This delayed transition is the entanglement stage: the model requires an extended
 280 training interval before it assigns sufficiently large probability to the optimal reasoning pattern.

281 Sec 6 provides additional case studies and numerical simulations that instantiate these two regimes.

282 5.4 Theoretical Explanation for RLIF

283 Proposition 5.2 shows that RLIF depends only on confidence in the final-answer distribution and
 284 therefore does not distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. This property
 285 explains why RLIF can ultimately degrade performance relative to the base model, consistent with
 286 the empirical results. At the same time, prior work reports that RLIF can improve model performance
 287 without external rewards Agarwal et al. [4], Zhao et al. [47]. We therefore analyze why RLIF can
 288 increase the accuracy of a well-trained LLM at the beginning of training.

289 We first impose the following condition on the base model:

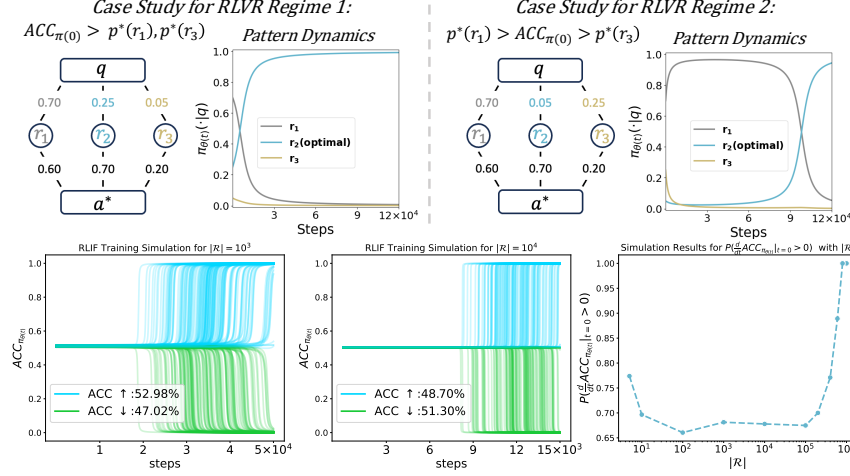


Figure 2: RLVR case studies (upper panels) and RLIF training simulations (lower panels). The RLVR panels instantiate two training regimes: rapid convergence to the optimal pattern (left) and an entanglement phase before convergence (right). The RLIF panels report the probability of convergence to different terminal states (left) and the probability of initial performance improvement (right) as $|\mathcal{R}|$ varies.

Assumption 5.5. For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$.

Because majority voting [35] is an effective inference-time improvement method, a well-trained LLM often assigns larger marginal probability to the correct answer $\mathbf{a}^*$ than to any incorrect answer; Assumption 5.5 is therefore plausible for modern LLMs. Under this condition, we analyze the binary-answer case $|\mathcal{A}| = 2$, assume that the success rate for each reasoning path is uniformly distributed, and consider high-entropy reasoning-pattern selection². We then characterize the derivative of overall accuracy at $t = 0$:

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:

1. The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.
2. With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.

The first statement in Theorem 5.6 establishes that the accuracy derivative is positive at initialization, providing a formal explanation for the early performance gains observed under RLIF. However, Proposition 5.2 implies that, when $\beta = 0$, continued optimization drives the policy deterministically toward the pattern $\mathbf{r}$ that maximizes $R_{\text{RLIF}}(\mathbf{r})$. The second statement in Theorem 5.6 shows that this maximizing pattern coincides with the minimum-success pattern with probability 0.5: $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$. Thus, even though RLIF initially increases accuracy, continued training can concentrate probability mass on reasoning paths with very low success rates, yielding final performance below that of the base model.

6 Case Studies and Numerical Simulations

To validate and interpret the theoretical results, we present case studies and numerical simulations corresponding to the preceding theorems. For RLVR, Fig. 2 (upper) illustrates the two training regimes characterized in Theorems 5.3 and 5.4. For RLIF, Fig. 2 (bottom) reports simulations in which each reasoning path’s success rate is sampled from a uniform distribution, thereby testing the predicted behavior at initialization and at convergence.

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

6.1 Case Studies for RLVR

Section 5.3 identifies two regimes in the RLVR gradient-flow dynamics. In the first regime (Eq 5.7), the selection probability of the optimal reasoning pattern, $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$, increases monotonically until convergence. In the second regime (Eq 5.8), the model first enters an entanglement phase and requires a substantially longer transition period before it follows the first-regime dynamics and converges to $\mathbf{r}^*$. We instantiate both regimes as follows:

- Fig. 2 (upper left) shows the favorable case in which the reference model satisfies Eq 5.7: its overall accuracy exceeds the success rate of every non-optimal reasoning pattern. Under this condition, probability mass rapidly concentrates on the optimal pattern.
- Fig. 2 (upper right) corresponds to Theorem 5.4. When a suboptimal pattern $\mathbf{r}'$ has a higher success rate than the initial overall accuracy, the model first remains in an entanglement phase. The transition time T_0 defined in Theorem 5.4 must elapse before the model enters the rapid-convergence regime. The displayed setting has large $\gamma_{\pi_{\text{ref}}}$ as defined in Eq 5.9: specifically, $\gamma_{\pi_{\text{ref}}} = 6$ because $(0.25 + 0.05)/0.05 = 6$. Since T_0 scales with $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$, this configuration substantially delays convergence to the optimal pattern. The observed pattern dynamics match the predicted entanglement behavior.

6.2 Numerical Simulations For RLIF

Section 5.4 shows that, under Assumption 5.5 and the base-model distribution specified in Theorem 5.6, the accuracy derivative at $t = 0$ is positive with probability 1 as $|\mathcal{R}| \rightarrow \infty$, while the policy has probability 50% of converging to the least accurate reasoning path. The simulations confirm both statements. Fig. 2 (bottom left) reports RLIF training outcomes across values of $|\mathcal{R}|$ and shows that convergence to an improved state ($ACC \uparrow$) occurs with approximately the same probability as convergence to a degraded state ($ACC \downarrow$), consistent with the second statement of Theorem 5.6. Fig. 2 (bottom right) evaluates the initial training step and shows that, for large $|\mathcal{R}|$, the probability of an initial performance improvement approaches 1, supporting the first statement of Theorem 5.6.

7 Conclusions and Limitations

This work provides an empirical and theoretical characterization of reinforcement-learning dynamics in LLM post-training. The analysis models reasoning as selection over latent reasoning patterns followed by answer generation, and it shows that RL primarily changes the distribution over these patterns rather than substantially changing each pattern’s intrinsic success rate. Within this framework, we derive training-dynamic results for two representative reward mechanisms, RLVR and RLIF, and validate the predicted regimes through case studies and numerical simulations. The study also has limitations that motivate future work: the semantic validity and interpretability of LLM-extracted reasoning patterns require further validation, the theoretical abstraction should be extended to more complex real-world reasoning processes, and the assumptions on the base-model distribution should be relaxed to cover broader policy classes.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

Section 4.2 analyzes reasoning-pattern dynamics on Number Theory tasks sampled from MATH. We additionally perform the same dynamic analysis on Geometry tasks, with results shown in Figure 3. Under RLVR, the model progressively increases probability mass on higher-accuracy reasoning patterns; under RLIF, pattern selection remains unstable. This behavior is consistent with the empirical observations and theoretical predictions in the main paper.

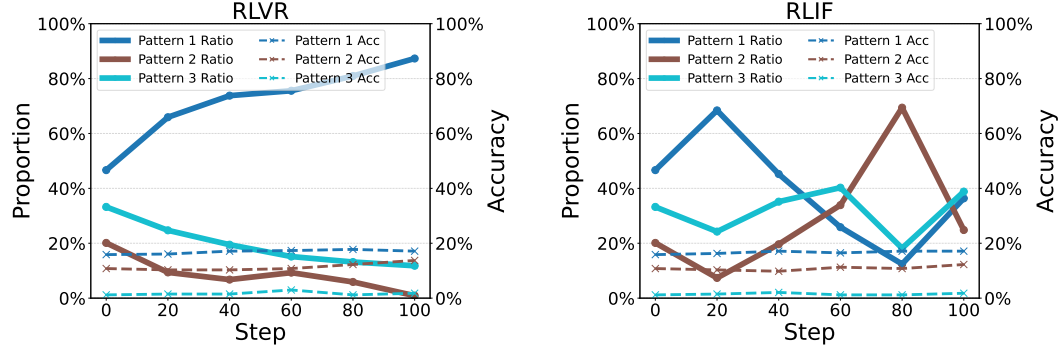


Figure 3: Dynamics of reasoning-pattern distributions and corresponding pattern accuracies for RL-enhanced models on Geometry tasks. Under RLVR, the model tends to select the highest-accuracy pattern (Pattern 1), whereas RLIF produces unstable pattern-selection behavior.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To test whether the findings transfer across model instances, we conduct supplementary experiments that evaluate the same empirical mechanism and theoretical predictions (Figure 4). We use Qwen-2.5-7B-Instruct [39] as the base model and compare it with Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source RLVR-enhanced model for mathematical reasoning. The evaluation covers number theory, geometry, algebra, calculus, and counting-and-probability tasks sampled from the MATH dataset [16], and also includes a complex problem from AMC23 [1]. Across these tasks, RLVR enhancement increases the frequency of high-accuracy reasoning patterns and decreases the frequency of less effective patterns. Appendix B.3 provides task-specific experimental details.

Across tasks, reasoning-pattern analysis shows that the highest-success patterns consistently become more frequent after RLVR enhancement, reinforcing the findings of Section 4 and Section 5. In Algebra and AMC23-19, shown in Figure 4, individual pattern success rates remain comparatively stable, which is consistent with the main empirical observations and provides additional support for Assumption 5.1.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To further evaluate the theory at larger scale, we conduct experiments with QwQ-32B [32], an RLVR-enhanced model based on Qwen-2.5-32B-Instruct [39]. We compare reasoning-pattern shifts between these two models on four tasks: Number Theory and Geometry from the MATH dataset [16], one task from AIME 2024 [7], and one task from OlympiadBench [15].

Most reasoning patterns produced by QwQ-32B correspond to the most accurate patterns identified for Qwen-2.5-32B, which is consistent with the theoretical prediction that RLVR increases mass on high-success patterns. We do not report QwQ pattern accuracies because the experiments use model APIs and therefore do not provide access to the CoT data required for the base model to continue generation from fixed partial reasoning traces. Consequently, pattern-level accuracy, as distinct from overall model accuracy, cannot be computed in this setting. Appendix B.2 describes the required “Accuracy Analysis” procedure.

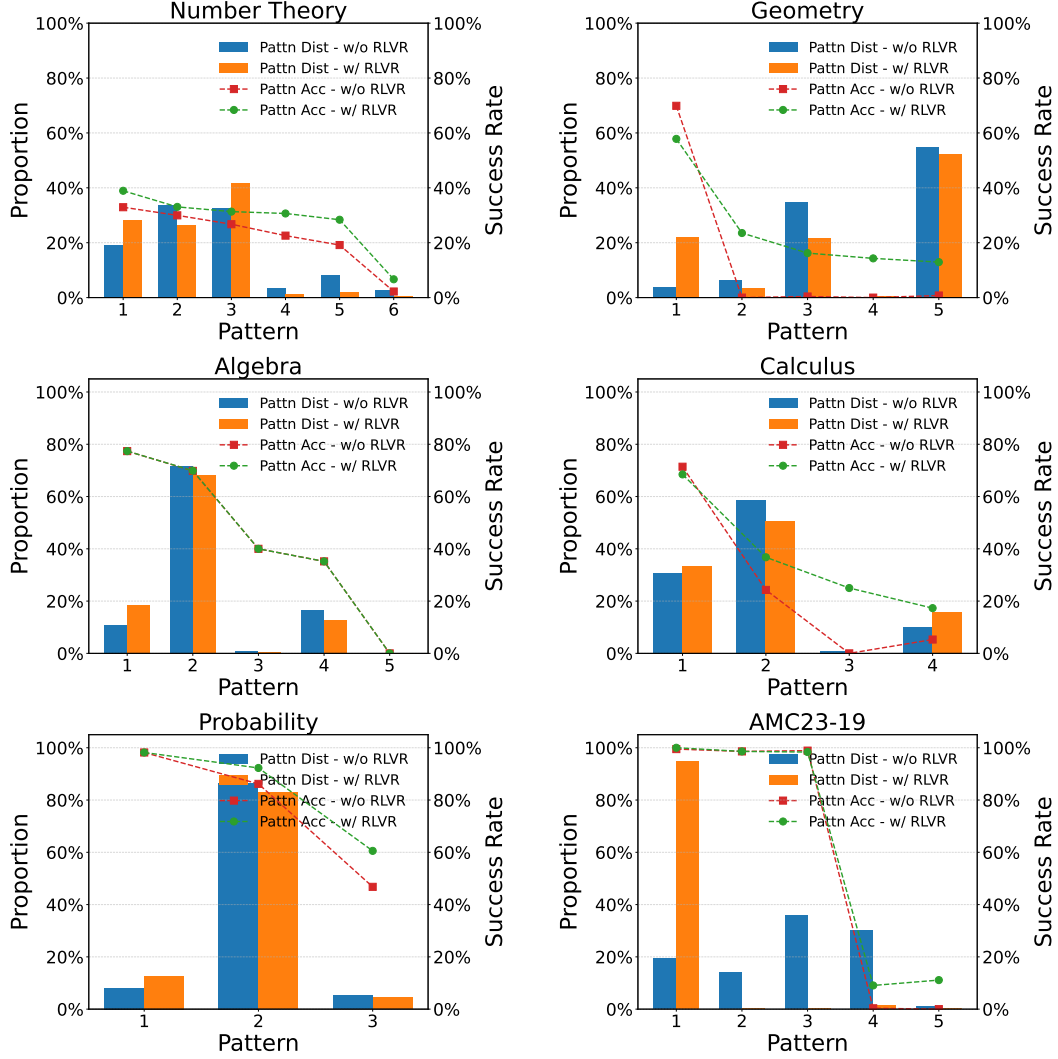
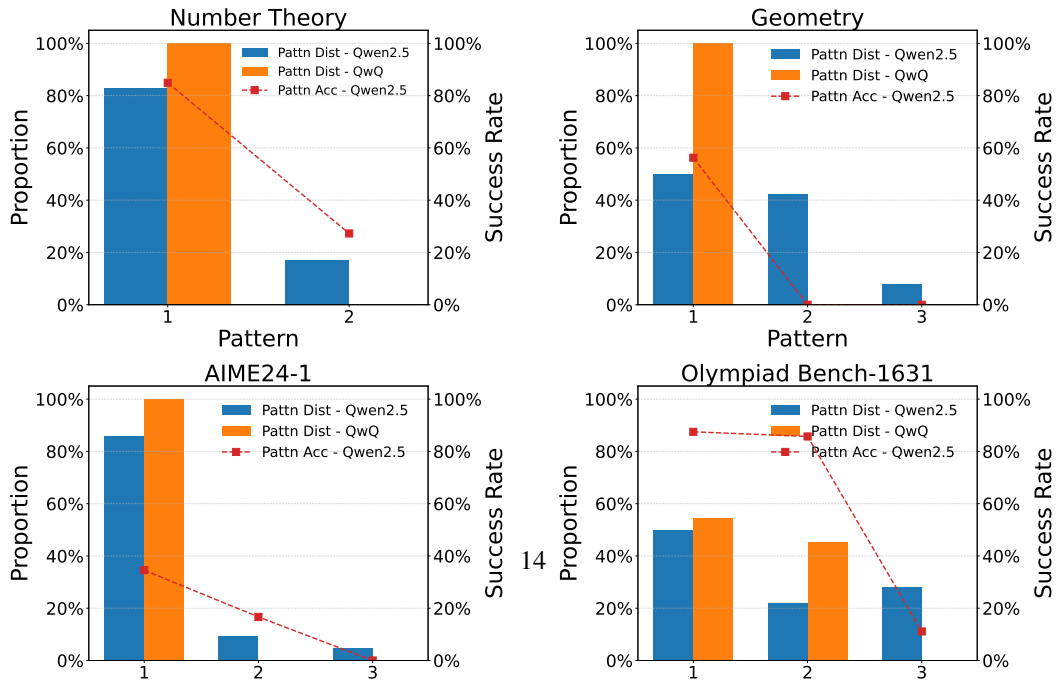


Figure 4: Reasoning-pattern distributions and corresponding success rates for models with and without RLVR enhancement across additional tasks. Bars (Pattern Dist) report each pattern’s proportion among all identified patterns, and dots (Pattern Acc) report the corresponding pattern accuracy or success rate. The results align with the empirical mechanism and theoretical conclusions.



B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: We use Qwen2.5-3B as the base model and train it on the MATH dataset under both RLVR and RLIF. All training runs use the verl [31] framework, with parameter settings matched to the MATH example provided by verl. For RLIF, the only modification is the reward-computation procedure. We run each method with three random seeds to obtain three training curves per reward mechanism.

Token-level Analysis: This experiment uses Qwen2.5-3B and the RL-enhanced models described above. For GSM8K and MATH, we randomly sample 400 questions from each test set; for AIME24, we evaluate the full dataset. For each question, we first sample an answer from the base model. We then concatenate that answer to the original question and provide the resulting sequence to the RL-enhanced model as a fixed prompt. Using the OpenAI library interface, we compute each token’s probability and its rank among all vocabulary-token probabilities at the same position. Comparing token ranks before and after RL gives the proportion of tokens whose ranks change.

Reasoning-pattern Level Analysis: We conduct two forms of reasoning-pattern analysis. First, using Qwen2.5-3B and its RL-enhanced checkpoints, we track pattern-distribution dynamics during training and evaluate every 20 steps. Second, to test whether the conclusions transfer to other base models and larger models, we evaluate Qwen-2.5-7B-SimpleRL-Zoo, QwQ-32B, and their corresponding base models. For these externally released models, intermediate checkpoints are unavailable, so we analyze only the pre- and post-RL pattern distributions rather than full training dynamics.

B.2 Reasoning Patterns Analysis Pipeline

Model Inference: We randomly sample questions across task types, such as Geometry and Algebra, and across difficulty levels, such as Level 4 and Level 5, from datasets including MATH. For each question in the pattern-distribution dynamics experiment, the model samples 1024 answers. For experiments that compare only pre- and post-RL distributions, the number of sampled answers ranges from 64 to 1024 depending on the task category. These responses constitute the input pool for subsequent pattern extraction and classification.

Reasoning Patterns Extraction: We sample a subset of base-model responses and use the GPT-4o [19] API to infer reasoning-pattern categories from those samples. Appendix D.1 provides the full extraction prompt. To make the extraction procedure deterministic, we set the GPT-4o API temperature to 0.

Responses Classification: We then use GPT-4o to assign every response to one of the extracted pattern categories, again with temperature set to 0. Appendix D.2 provides the full classification prompt.

Accuracy Analysis: To isolate the effect of pattern-distribution changes on accuracy, we construct prompts that concatenate the original question with a partial base-model response representing a specific reasoning pattern. This intervention fixes the observed reasoning process and allows us to test whether accuracy changes are attributable to the distribution over patterns rather than to changes within a pattern. Concretely, we remove the sentence containing the final answer from the base-model response, append the preceding reasoning text to the question, and use this concatenated sequence as the prompt for the RL-enhanced model. The accuracy of the continuation generated by the RL-enhanced model is treated as the accuracy of the corresponding reasoning pattern.

B.3 Details for Each Task

We next specify the experimental procedure for each task, including task definitions and representative reasoning-pattern examples (Table 2). Appendix D.3 provides example task prompts.

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Representative reasoning-pattern examples across tasks.

Number Theory: This task asks the model to solve coin-redistribution problems across multiple bags. The initial state contains equal numbers of coins in each bag; after receiving additional coins and redistributing all coins equally, the total number of coins must exceed a specified threshold. The model must determine the minimum feasible initial coin count per bag. We derive the task from a level 5 MATH problem (id: test/number_theory/1055) [16] and construct 32 evaluation variants by modifying parameters such as the number of bags and coin totals. We use one-shot prompting with a simple unrelated example to enforce the "boxed" answer format without introducing task-relevant reasoning patterns. For evaluation, we select 20 questions and 4 base-model responses per question for pattern extraction. We then sample 64 responses per question and classify all responses according to the extracted patterns, enabling comparison of reasoning-pattern distributions before and after RL.

Geometry: For the geometry task, the model receives the lengths of two sides of a scalene triangle and must determine how many integer centimeter values are possible for the third side. The task is derived from a level 4 MATH geometry problem (id: test/geometry/1046). The evaluation pipeline matches the Number Theory setting: we construct a synthetic dataset of 32 questions, sample 20 questions with 4 base-model responses each from Qwen2.5-7b-Instruct for pattern extraction, and then compare models with and without RLVR enhancement using 64 responses per question over the full evaluation set.

Algebra: The algebra task is a can-recycling problem. Given an initial number of old cans and a conversion rate in which n old cans produce one new can, the model must compute the total number of new cans that can eventually be produced. The difficulty arises because each iterative recycling step may leave remainders that later combine with newly produced cans. The task template is taken from

a level 4 MATH algebra problem (id: test/algebra/2768). For this and the following supplementary tasks, we use the same dataset size, sampling counts, and analysis protocol as in the preceding tasks.

Calculus: The calculus task presents a single-variable trigonometric equation together with the variable domain and asks the model to count the number of solutions. Although the task is relatively simple, it probes basic symbolic reasoning for pre-calculus expressions. The prototype is a level 4 MATH pre-calculus problem (id: test/precalculus/1140).

Counting and Probability: The counting-and-probability task specifies a bin containing white and black balls. Two balls are drawn at random, the probability of drawing one ball of each color is provided, and the number of balls of one color is fixed. The model must compute the minimum feasible number of balls of the other color. This task is derived from a level 4 MATH counting-and-probability problem (id: test/counting_and_probability/79).

AMC23 Question: We use question 19 from AMC23 [1]. For reasoning-pattern extraction, we randomly sample 48 responses; for reasoning-pattern analysis, we sample 1024 responses.

B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

We evaluate four tasks in the larger-model setting. Because testing is API-limited, we randomly select one question per task, sample 64 responses from each of the two models, and perform reasoning-pattern extraction and classification directly on these answers. For the MATH tasks, we use question 1055 from Number Theory and question 1046 from Geometry. For the remaining tasks, we use problem 1 from AIME24 and question 1631 from OlympiadBench. Table 2 provides representative pattern examples.

C Deferred Proofs

This appendix provides the deferred proofs for the main theoretical results: Proposition 5.2, Theorem 5.3, Theorem 5.4, and Theorem 5.6.

C.1 Proof of Proposition 5.2

Proof of Proposition 5.2. The proof follows the KL-regularized optimization argument of [26]. Recall that the RL objective is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

Under the framework in Section 5.1, the policy first samples a reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ according to $\pi_{\theta}(\mathbf{r}_i|\mathbf{q})$ and then samples the final answer $\mathbf{a} \in \mathcal{A}$ according to $\pi_{\theta}(\mathbf{a}|\mathbf{r}_i)$. With this factorization, the reward objective can be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

Define $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$. Then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot|\mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[\ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

635 The third equality uses Assumption 5.1. The quantity $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} p^*(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the
 636 partition constant that ensures

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta}(\mathbf{r}|\mathbf{q}),$$

637 is a valid probability distribution satisfying $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$; Z is constant with respect to $\mathbf{r}$.

638 Therefore, maximizing the objective in Equation C.2 is equivalent to the following KL minimization
 639 problem:

$$\begin{aligned} & \max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ & = \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

640 By nonnegativity of KL divergence, the optimal policy for the KL-regularized reward-maximization
 641 objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

642 For RLVR, this gives $R_{\text{RLVR}}(\mathbf{r}) = p^*$. For RLIF, we obtain

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \log|\mathcal{A}| \right)}_c, \end{aligned}$$

643 where the second term is identical for all $\mathbf{r} \in |\mathcal{R}|$. Hence, for RLIF, the optimal policy can be written
 644 as

$$\begin{aligned} \pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \end{aligned}$$

645 So we can write $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r}))$. This concludes the proof
 646 of the theorem. $\square$

647 C.2 Dynamics of Tabular Policy

648 Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the
 649 gradient dynamics of θ can be computed as

$$\begin{aligned} \frac{d}{dt} \theta(t) &= \nabla \phi_{\text{RL}}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})], \end{aligned}$$

Where $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\boldsymbol{\theta}_{:,q})_{\mathbf{r}}$, $\boldsymbol{\theta}_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of $\boldsymbol{\theta}$ corresponding to $\mathbf{q}$, and $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned} \frac{\partial \boldsymbol{\theta}_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) \left(\nabla \text{softmax}(\boldsymbol{\theta}_{:,q})_{\mathbf{r}_j} \right)_{\mathbf{r}_i} \right] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) \left(-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \right) \right] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \end{aligned}$$

where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \boldsymbol{\theta}_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \boldsymbol{\theta}_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t}$$

C.3 Proof of Theorem 5.3

To proof Theorem 5.3, here we first prove that $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4), then we establish a lower bound for $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$ (Eq C.5), finally we derive the final bound for t stated in Theorem 5.3.

Proof of Theorem 5.3. Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial \boldsymbol{\theta}_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \boldsymbol{\theta}_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \boldsymbol{\theta}_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}), \end{aligned}$$

and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) p^*(\mathbf{r}_i) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \quad (\text{C.4}) \end{aligned}$$

As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$

663 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\
&\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\
&= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
&\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \geq 0. \tag{C.5}
\end{aligned}$$

664 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})$ holds for $t \geq 0$, define $C =$
665 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^*|\mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.$$

666 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})}}.$$

667 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \epsilon.$$

668 This concludes the proof of the theorem. $\square$

669 C.4 Proof of Theorem 5.4

670 To proof Theorem 5.4, we first establish an upper bound for $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ (Eq C.6), then we analyze
671 the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$ (Eq C.7) and prove that for $t > T_0$, $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$.

672 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
673 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
&= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
&\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}')) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
&\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
&\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
\end{aligned}$$

674 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \tag{C.6}$$

675 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned} \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\ &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0 \end{aligned}$$

676 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 677 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

678 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned} \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}} \end{aligned}$$

679 Then we have:

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

680 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

681 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

682 Let $C_2 = 1/\Delta$, $C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

683 then:

$$\begin{aligned}
& \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\
& \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\
& \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta}\right)^{2\gamma_{\pi_{\text{ref}}}/\Delta}\right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\
& = \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta}\right)^{2\gamma_{\pi_{\text{ref}}}/\Delta}\right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\
& = \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}').
\end{aligned}$$

684 Finally, for $t > T_0$ we have:

$$\begin{aligned}
\text{ACC}_{\theta(t)} - p^*(\mathbf{r}') & \geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\
& \geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0.
\end{aligned}$$

685 This concludes the proof of the theorem. $\square$

686 C.5 Proof of Theorem 5.6

687 *Proof of Theorem 5.6.* First consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$. Under Assumption
688 5.5, this implies $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of the overall accuracy is

$$\begin{aligned}
\frac{d}{dt} \text{ACC}_{\theta(t)} & = \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\
& = \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) - \text{Con}_{\theta(t)} \right),
\end{aligned}$$

689 where $\text{Con}_{\theta(t)} = - \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q}))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
690 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned}
\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} & = \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\
& \quad \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_j, \mathbf{q})) \right) \\
& = \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) \right).
\end{aligned}$$

691 We consider the limit $|\mathcal{R}| \rightarrow +\infty$ with success rates $p^*(\mathbf{r})$ sampled i.i.d. from $U[0, 1]$. By the central
692 limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r}))$ concentrates around its expectation
693 0.5 with deviation order $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. Define this positive deviation as $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ such
694 that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$.

695 According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$,
696 the empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$

converges to the density: $f(x) = 1 + 12\delta(x - 0.5)$, $x \in [0, 1]$. As $|\mathcal{R}| \rightarrow \infty$, we can approximate the summation over $\mathbf{r}$ with an integral over this limiting distribution $f(x)$. The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 -(1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3}\delta > 0$$

Since $\delta > 0$, the derivative is positive. In the limit $|\mathcal{R}| \rightarrow \infty$, this event holds with probability 1, proving the first statement.

To prove the second statement, we analyze the RLIF reward function. For an answer space of size $|\mathcal{A}| = 2$, the reward assigned to a path $\mathbf{r}$ is proportional to the negative log-likelihood under the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | q, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing $R_{\text{RLIF}}(\mathbf{r})$ is equivalent to selecting the path whose success rate $p^*(\mathbf{r})$ is farthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The highest-reward path has success rate $p_{\min}^*$ precisely when this value is farther from 0.5 than $p_{\max}^*$ is. Equivalently:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to $p_{\min}^* + p_{\max}^* < 1$.

We now compute the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting density $f(x) = 1 + 12\delta(x - 0.5)$. By standard order-statistic limits, the scaled minimum and upper-tail maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables. Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) = 1 - 6\delta$.
- Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) = 1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be expressed in terms of these asymptotic variables as:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this probability is:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, we have $\delta \rightarrow 0$ as $|\mathcal{R}| \rightarrow \infty$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

This concludes the proof of the second statement.

□

D Full Prompts

D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

Example processes from input

Common elements

...

There are reasoning processes:

{extracted_responses}

Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

718

719 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.

{patterns_categories}

Here are the reasoning processes to classify:

{extracted_responses}

Please output your classification in this JSON format:

720

721 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

722 <lim_start>system
You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>
<lim_start>
user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_end>
<lim_start>assistant
Let me solve this step by step.
<think>

723

Prompt Example for Long Arithmetic

<lim_start>system
You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>
<lim_start>user
Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of $a + b = c$ or $a - b = c$, for example " $1 + 2 = 3$ ".<lim_end>
<lim_start>assistant
Let me solve this step by step.
<think>

724

Prompt Example for Number Theory

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within \boxed{ }.<lim_end>
<lim_start>assistant

725

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4\$ centimeters and 7\$ centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within \boxed{ }.<lim_end>
<lim_start>assistant

726

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

727

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

728

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

729

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

730

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a 9 -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

731

Prompt Example for OlympiadBench-Q1631

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

732